MACHINE LEARNING

PRODUCT DEMAND FORECAST USING RIDGE AND LASSO REGRESSION

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PROBLEM STATEMENT

Businesses often face difficulty in accurately estimating future product demand. Overestimating demand can result in excess inventory and increased storage costs, while underestimating demand can cause stockouts and lost sales. This project aims to develop a machine learning system that uses historical sales and relevant store factors to forecast future demand and support better inventory and business planning.

OBJECTIVES

1. Analyze historical retail sales data.
2. Preprocess and transform the data for machine learning.
3. Engineer features that capture sales and seasonal patterns.
4. Develop Ridge and Lasso Regression models.
5. Compare their performance using regression metrics.
6. Predict future sales/demand for new data.
7. Provide useful predictions for inventory and sales planning.

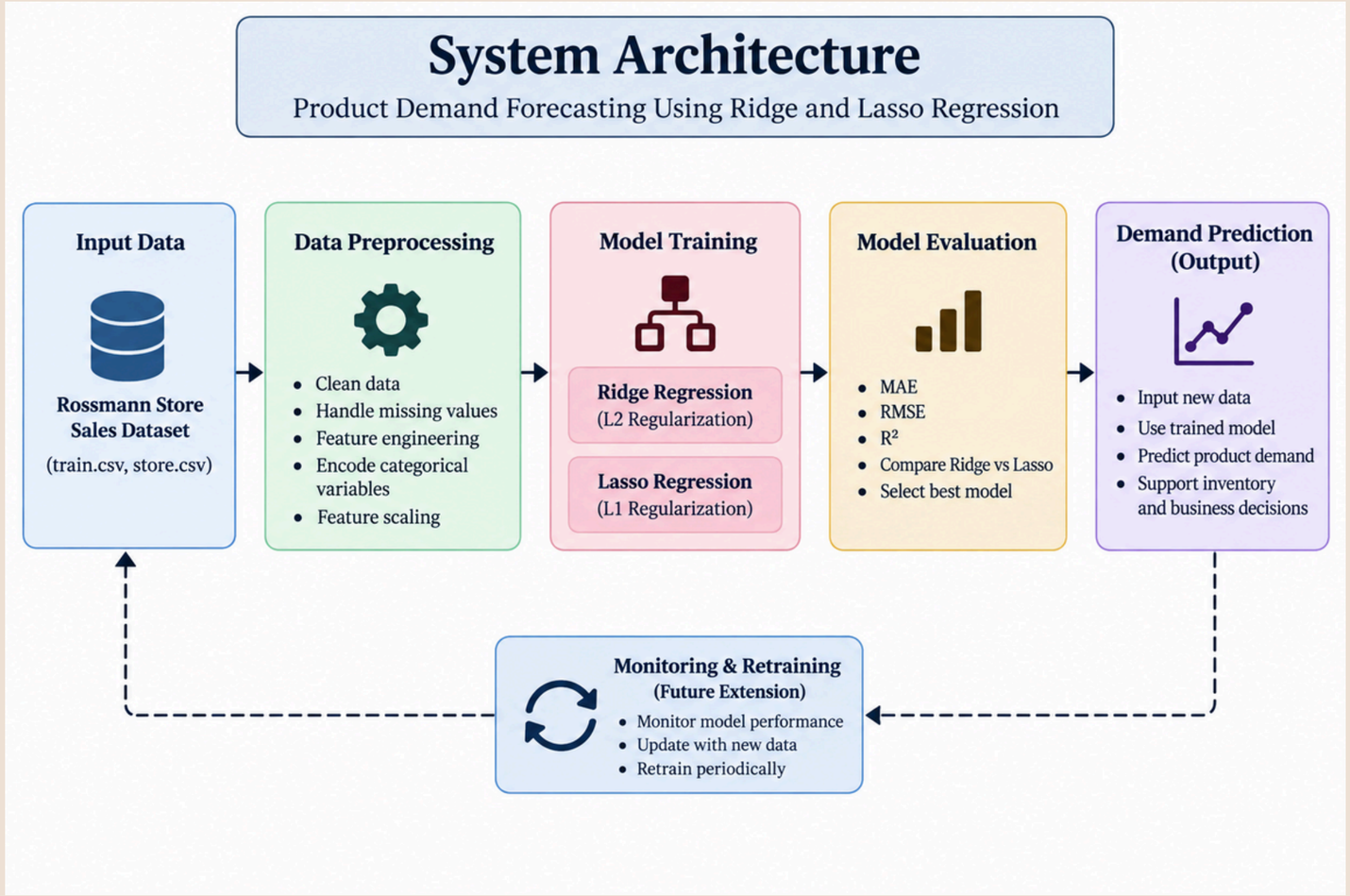
DATASET

Rossmann Store Sales Dataset

Key Features Used in This Project

Feature Name	Data Type	Description
Sales	Integer	Total sales for the day **(target variable)**
Customers	Integer	Number of customers in the store
Promo	Integer	Indicates whether a promotion was running (1 = yes, 0 = no)
StateHoliday	Categorical	State holiday indicator (a = public holiday, b = Easter holiday, c = Christmas, 0 = none)
SchoolHoliday	Integer	Indicates whether it was a school holiday (1 = yes, 0 = no)
StoreType	Categorical	Type of store (a, b, c, d)
Assortment	Categorical	Assortment level (a = basic, b = extra, c = extended)
CompetitionDistance	Float	Distance to the nearest competitor store (in meters)
Promo2	Integer	Indicates participation in a continued promotion (1 = yes, 0 = no)
DayOfWeek	Integer	Day of the week (1 = Monday, …, 7 = Sunday)
Date	Date	Date of the record (used for extracting time-based features such as month, year, etc.)
Open	Integer	Indicates whether the store was open (1 = open, 0 = closed)

SYSTEM ARCHITECTURE



II

CONCLUSION

The project aims to develop a machine learning system for product demand forecasting using historical retail sales data. Ridge and Lasso Regression will be used to predict future demand while reducing overfitting and identifying relevant features. The models will be evaluated using MAE, RMSE, and R^2 to determine their performance. The final system will provide demand predictions that can support inventory planning, sales forecasting, stock management, and better business decision-making.

FUTURE SCOPE

The system can be further enhanced by incorporating real-time sales data and advanced machine learning models such as Random Forest, XGBoost, and Gradient Boosting for improved demand forecasting. Future development can include a real-time prediction API and an interactive dashboard to visualize sales trends and predicted demand. Model monitoring and automatic retraining can also be implemented to adapt to changing demand patterns. The system can eventually be extended to provide automated inventory and stock-replenishment recommendations based on predicted demand.

Thank you